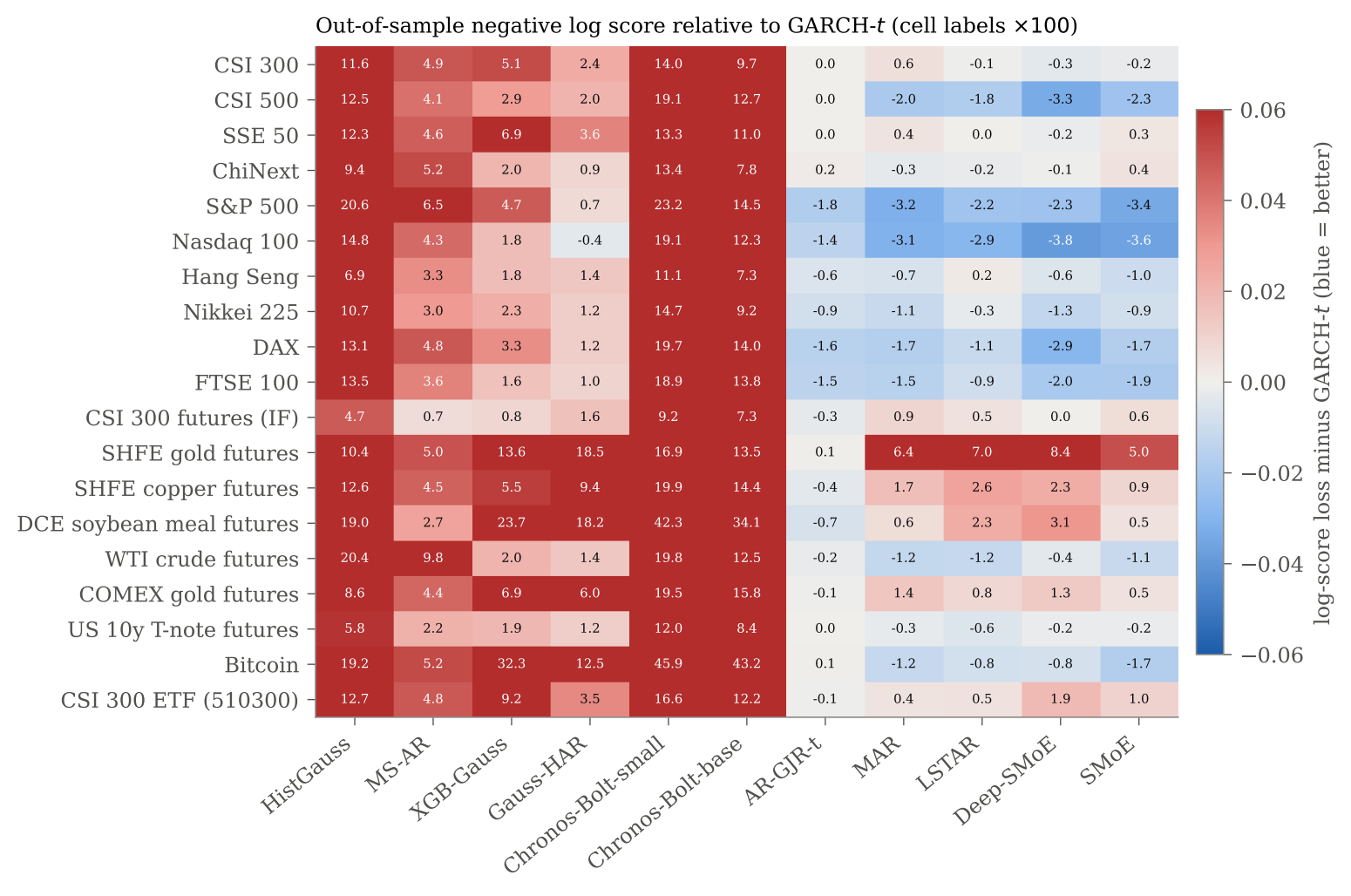

Out-of-sample negative log score relative to GARCH-t (cell labels ×100)
CSI 300 11.6 4.9 5.1 2.4 14.0 9.7 0.0 0.6 -0.1 -0.3 -0.2
CSI 500 12.5 4.1 2.9 2.0 19.1 12.7 0.0 -2.0 -1.8 -3.3 -2.3
SSE 50 12.3 4.6 6.9 3.6 13.3 11.0 0.0 0.4 0.0 -0.2 0.3
ChiNext 9.4 5.2 2.0 0.9 13.4 7.8 0.2 -0.3 -0.2 -0.1 0.4
S&P 500 20.6 6.5 4.7 0.7 23.2 14.5 -1.8 -3.2 -2.2 -2.3 -3.4
Nasdaq 100 14.8 4.3 1.8 -0.4 19.1 12.3 -1.4 -3.1 -2.9 -3.8 -3.6
Hang Seng 6.9 3.3 1.8 1.4 11.1 7.3 -0.6 -0.7 0.2 -0.6 -1.0
Nikkei 225 10.7 3.0 2.3 1.2 14.7 9.2 -0.9 -1.1 -0.3 -1.3 -0.9
DAX 13.1 4.8 3.3 1.2 19.7 14.0 -1.6 -1.7 -1.1 -2.9 -1.7
FTSE 100 13.5 3.6 1.6 1.0 18.9 13.8 -1.5 -1.5 -0.9 -2.0 -1.9
CSI 300 futures (IF) 4.7 0.7 0.8 1.6 9.2 7.3 -0.3 0.9 0.5 0.0 0.6
SHFE gold futures 10.4 5.0 13.6 18.5 16.9 13.5 0.1 6.4 7.0 8.4 5.0
SHFE copper futures 12.6 4.5 5.5 9.4 19.9 14.4 -0.4 1.7 2.6 2.3 0.9
DCE soybean meal futures 19.0 2.7 23.7 18.2 42.3 34.1 -0.7 0.6 2.3 3.1 0.5
WTI crude futures 20.4 9.8 2.0 1.4 19.8 12.5 -0.2 -1.2 -1.2 -0.4 -1.1
COMEX gold futures 8.6 4.4 6.9 6.0 19.5 15.8 -0.1 1.4 0.8 1.3 0.5
US 10y T-note futures 5.8 2.2 1.9 1.2 12.0 8.4 0.0 -0.3 -0.6 -0.2 -0.2
Bitcoin 19.2 5.2 32.3 12.5 45.9 43.2 0.1 -1.2 -0.8 -0.8 -1.7
CSI 300 ETF (510300) 12.7 4.8 9.2 3.5 16.6 12.2 -0.1 0.4 0.5 1.9 1.0
HistGauss
MS-AR
XGB-Gauss
Gauss-HAR
Chronos-Bolt-small
Chronos-Bolt-base
AR-GJR-t
MAR
LSTAR
Deep-SMoE
SMoE
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
log-score loss minus GARCH-t (blue = better)